

Data-Driven SoC Floorplanning

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Abstract—System-on-Chip (SoC) floorplanning requires placing blocks on a 2D canvas to jointly minimize wirelength and bounding-box area while satisfying hard constraints (non-overlap, area/shape limits, immutability) and soft constraints (grouping, multi-instantiation, and boundary preferences). Traditional search-based methods such as simulated annealing scale poorly on large designs. We instead cast floorplanning as conditional generation and amortize the search with *diffusion models*, trained on the FloorSet-Lite benchmark whose one million optimal-by-construction layouts let us measure the optimality gap directly. We develop and compare two diffusion models that differ in how they encode block features, connectivity, and constraints: Model A, a Transformer denoiser with connectivity-biased attention, and Model B, a graph neural network augmented with step-wise spatial edges and a constraint-aware inference stack. Both decode an area-exact continuous layout, which a shared legalization stage then drives to exactly zero overlap for feasibility. On FloorSet-Lite instances of 21–120 blocks, the GNN model lowers residual pre-legalization overlap by over an order of magnitude (55.6 vs. 1455.5) at a comparable area gap, while staying well within the 60s per-instance budget. A shared constraint-graph minimum-displacement legalizer then drives every instance to exactly zero overlap—raising feasibility to 100%—while keeping preplaced and fixed-shape blocks immutable and adding only a few seconds per instance; because it perturbs the layout minimally, the GNN model’s order-of-magnitude lower residual overlap translates into smaller wirelength and area damage during legalization.

I. INTRODUCTION

Modern System-on-Chip (SoC) floorplanning is a complex, multi-objective NP-complete optimization problem. Unlike academic settings, blocks must be arranged on a 2D canvas to jointly minimize total wirelength and bounding-box area while strictly satisfying hard constraints (non-overlap, area/shape limits) and accommodating soft constraints (grouping, multi-instantiation, and boundary preferences).

SoC floorplanning extends conventional bin packing by introducing spatial dependencies: flexible-shape blocks with predefined area budgets, preplaced or fixed-shape blocks, and interconnected nets reaching external terminals. These dependencies directly drive total wirelength, requiring floorplanners to simultaneously optimize geometry, connectivity, and heterogeneous constraints.

Traditional search-based methods, such as simulated annealing, suffer from poor runtime scalability on large designs. This motivates our shift toward learning-based methods to amortize the search cost. In this work, we propose a machine-learning approach based on *diffusion models* to generate

feasible floorplans with significantly reduced runtimes. We train and evaluate our models on the FloorSet-Lite benchmark, leveraging its one million optimal-by-construction layouts to precisely measure the optimality gap.

We formulate floorplanning as a conditional generation problem. Given block specifications, connectivity, and constraints, our diffusion models generate a continuous layout representation, which is subsequently decoded and legalized to ensure zero overlap. Under this framework, we independently develop and compare two diffusion-based solutions using alternative conditioning encoders.

II. PROBLEM STATEMENT

A. Inputs

- **$B = \{b_1, \dots, b_k\}$** : Set of k soft blocks with predefined area targets $A = \{a_1, \dots, a_k\}$.
- **$T = \{t_1, \dots, t_r\}$** : Set of r fixed external terminals.
- **Connectivity Matrices**: $W^{(int)} \in \mathbb{R}^{k \times k}$ for inter-block nets, and $W^{(ext)} \in \mathbb{R}^{k \times r}$ for block-to-terminal nets.
- **Soft Constraints** (violations incur penalty costs):
 - **Grouping**: Specified subsets of blocks ($B_{grouping}^P$) must be physically abutted.
 - **Multi-Instantiation Blocks (MIB)**: Blocks within defined groups (B_{mib}^Q) must share identical dimensions.
 - **Boundary**: Designated blocks ($B_{boundary}$) must touch the specified bounding-box edges.
- **Hard Constraints** (violations render solutions infeasible):
 - **Area/Dimension Limits**: Soft blocks must satisfy target areas within a 1% tolerance:

$$\left| \frac{w_i h_i - a_i}{a_i} \right| \leq 0.01$$

- **Overlap-Free**: All block intersections must be exactly zero:

$$\text{Area}(b_i \cap b_j) = 0 \quad \forall i \neq j$$

- **Immutability**: Preplaced blocks ($B_{preplaced}$) must exactly retain input coordinates and dimensions. Fixed-shape blocks (B_{fixed}) must exactly retain their input dimensions.

B. Expected Output

- **Locations:** $L = \{(x_i, y_i) | i = 1, \dots, k\}$, representing the lower-left coordinates of each block.
- **Dimensions:** $D = \{(w_i, h_i) | i = 1, \dots, k\}$, representing width and height. Block b_i occupies the region $[x_i, x_i + w_i] \times [y_i, y_i + h_i]$.

C. Objective Function

We use the following multi-objective cost function:

$$\text{Cost} = \begin{cases} \left(1 + \alpha (\text{HPWL}_{\text{gap}} + \text{Area}_{\text{bbox}}^{\text{gap}})\right) \times e^{\beta \cdot \text{Viol}^{\text{rel}}} \times \max(0.7, \text{RF}^\gamma), & \text{if feasible} \\ M, & \text{if infeasible} \end{cases} \quad (1)$$

where:

- Infeasible solutions receive a fixed penalty $M = 10$. Feasible costs are capped at $M - 10^{-6}$.
- Weighting hyperparameters: $\alpha = 0.5$, $\beta = 2.0$, and $\gamma = 0.3$.
- HPWL^{gap} and $\text{Area}_{\text{bbox}}^{\text{gap}}$ are relative gaps to the optimal baseline wirelength and area.
- $\text{Viol}^{\text{rel}} \in [0, 1]$ quantifies soft-constraint violations.
- RF is the runtime normalized by the median submission runtime. The runtime benefit is capped at 30% ($\max(0.7, \cdot)$).

III. PROPOSED TECHNIQUE

Given the 1M data set (FloorSet), with the corresponding optimal solutions. We approach the task with two diffusion-based models that share the same problem formulation—casting floorplanning as conditional generation over a continuous layout representation—but differ in how the input is encoded. Model A uses an attention-based encoder, whereas Model B uses a graph neural network. Since the raw diffusion output may still contain residual block overlaps, a legalization step is applied to both models to remove overlaps and further reduce the cost.

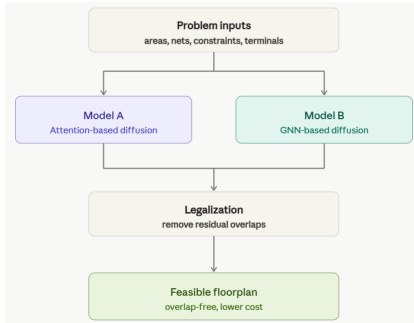


Fig. 1. Overview of the proposed pipeline. The same problem inputs are fed to two independent diffusion models—an attention-based model (Model A) and a GNN-based model (Model B). The raw output of each model may still contain residual block overlaps, so a shared legalization step is applied to remove overlaps and obtain a feasible floorplan.

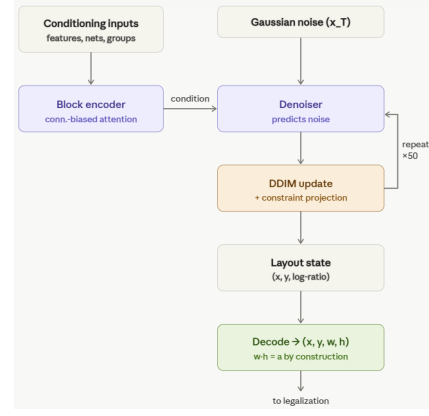


Fig. 2. Internal pipeline of Model A. A block encoder with connectivity-biased attention produces a conditioning embedding; starting from Gaussian noise, the denoiser and a constraint-aware DDIM update are iterated to a layout state, which is decoded into block coordinates and dimensions.

A. Model A: Attention-based diffusion

1) **Overview:** Model A is a Transformer-based diffusion model (Fig. 2). Each block is represented by its normalized position and a log aspect ratio, from which width and height are recovered analytically, so the area constraint is satisfied by construction. A *block encoder* encodes the block features, connectivity, and group affinities into a conditioning embedding; both the encoder and the *denoiser* use a *connectivity-biased attention* mechanism that injects connectivity and group membership as learnable biases. Starting from Gaussian noise, the denoiser and a DDIM update iteratively reconstruct a layout state while hard constraints are enforced during sampling; the final state is decoded and passed to the shared legalization stage.

2) **Floorplan Representation:** Each block b_i is represented by a continuous state $s_i = (\tilde{x}_i, \tilde{y}_i, \rho_i)$, where $\tilde{x}_i, \tilde{y}_i \in [0, 1]$ are the normalized lower-left coordinates and $\rho_i = \log(w_i/h_i)$ is the log aspect ratio. Given the target area a_i , the dimensions are decoded as

$$w_i = \sqrt{a_i e^{\rho_i}}, \quad h_i = \sqrt{a_i e^{-\rho_i}}, \quad (2)$$

which guarantees $w_i h_i = a_i$ for any ρ_i ; the area constraint is thus satisfied by construction, and the model learns only positions and aspect ratios. The model is conditioned on per-block features $f_i \in \mathbb{R}^{12}$ (Table I), the inter-block connectivity matrix $W^{(int)}$, and a group-affinity matrix G .

3) **Diffusion Process:** The model generates layouts by reversing a noising process. In the forward process, Gaussian noise is gradually added to a clean layout x_0 , and the noisy layout at step t follows

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (3)$$

where x_0 stacks the ground-truth states $\{s_i\}$, $\bar{\alpha}_t = \prod_{\tau=1}^t (1 - \beta_\tau)$, and β_τ increases linearly from 10^{-4} to 2×10^{-2} . A Transformer denoiser ϵ_θ , taking the noisy state, a sinusoidal timestep embedding, and the conditioning embedding, is trained to predict the added noise.

TABLE I
PER-BLOCK FEATURE VECTOR $f_i \in \mathbb{R}^{12}$.

Index	Feature	Description
0	a_i	Normalized target area
1	$fixed$	Fixed-shape flag
2	$preplaced$	Preplaced flag
3	mib	MIB group membership
4	$group$	Grouping membership
5–8	$\beta_{L/R/T/B}$	Boundary edge bits
9	p_x	Weighted pin centroid (x)
10	p_y	Weighted pin centroid (y)
11	p_w	Normalized total pin weight

4) **Attention-based Conditioning:** Both the encoder and the denoiser use a *connectivity-biased attention* mechanism. Beyond the standard scaled dot-product term, the attention logits are augmented with two learnable biases:

$$A = \frac{QK^\top}{\sqrt{d_k}} + \phi(\log(1 + W^{(int)})) + \psi(G), \quad (4)$$

where ϕ, ψ are learnable per-head projections. The first term makes strongly connected blocks attend to each other, implicitly guiding wirelength minimization; the second uses $G_{ij} = 1$ for blocks in the same MIB or grouping constraint, helping the model coordinate their shapes or positions. Unlike a GNN, which treats the netlist as a hard structural prior, this injects connectivity as soft, learnable biases over a fully-connected attention map.

5) **Training Objective:** The network is trained to predict the noise added at each step, using a masked mean-squared error that counts only real blocks:

$$\mathcal{L}_{mse} = \mathbb{E}_{t, \epsilon} \left[\left\| m \odot (\epsilon - \epsilon_\theta(x_t, t, c)) \right\|^2 \right]. \quad (5)$$

As this alone does not reward constraint satisfaction, we recover the estimated clean layout $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta) / \sqrt{\bar{\alpha}_t}$ and add three differentiable penalties on it:

$$\mathcal{L}_{mib} = \sum_q \text{Var}_{i \in M_q}(\hat{\rho}_i), \quad (6)$$

$$\mathcal{L}_{grp} = \sum_p \text{Var}_{i \in G_p}(\hat{x}_i, \hat{y}_i), \quad (7)$$

$$\mathcal{L}_{bnd} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} d(s_i, \partial)^2. \quad (8)$$

$\mathcal{L}_{mib}$ is the variance of aspect ratios within a MIB group (small only when shapes match); $\mathcal{L}_{grp}$ is the spatial variance of block centers (pulling grouped blocks together); and $\mathcal{L}_{bnd}$ is the squared distance from a boundary-constrained block to its required edge. The total loss is

$$\mathcal{L} = \mathcal{L}_{mse} + \lambda_{mib} \mathcal{L}_{mib} + \lambda_{grp} \mathcal{L}_{grp} + \lambda_{bnd} \mathcal{L}_{bnd}. \quad (9)$$

The penalties are enabled only after a warm-up period, and an EMA of the weights is used at inference.

6) **Constraint-aware Sampling:** At inference we use DDIM sampling ($\eta = 0$), which is much faster than ancestral sampling and benefits the runtime objective. Hard constraints are enforced by inpainting: preplaced blocks are overwritten with their ground-truth state at each step, while for fixed-shape blocks only the aspect-ratio dimension is overwritten. To satisfy MIB constraints, during the final sampling steps the aspect ratios within each MIB group are projected onto their group mean.

B. Model B: GNN-based diffusion

1) **Overview:** Model B is a graph-neural-network diffusion model. Each block is a node whose continuous state encodes its center and aspect ratio; width and height are recovered analytically, so the area constraint holds by construction. Unlike Model A, which injects connectivity as soft attention biases over a fully-connected map, Model B encodes the netlist as an *explicit graph* and augments it with spatial k -nearest-neighbor edges rebuilt at every denoising step, giving the model a direct structural prior over both connectivity and local geometry. A message-passing denoiser, conditioned on a scalar solution-quality signal via FiLM, runs a DDIM reverse process equipped with a *constraint-aware inference stack* that resolves overlaps and soft constraints during sampling, followed by a force-directed spreading pass before the shared legalization stage.

2) **Floorplan Representation:** Each block b_i is represented by a continuous state $s_i = (\tilde{c}_i^x, \tilde{c}_i^y, \rho_i)$, where $\tilde{c}_i^x, \tilde{c}_i^y$ are the normalized block-center coordinates and $\rho_i = \log(w_i/h_i)$ is the log aspect ratio. As in Model A, the dimensions are decoded by Eq. (2), guaranteeing $w_i h_i = a_i$, so the model learns only positions and aspect ratios. Each node carries a 44-dimensional input feature: a sinusoidal positional encoding (32), the state (3), the target area (1), a preplaced flag (1), the weighted pin centroid and a has-pins flag (3), and the four boundary bits (4). Edges carry an 8-dimensional attribute vector

$$e_{ij} = [w_{ij}^{(int)}, \text{is-net}, \text{is-spatial-centroid}, \text{is-spatial-edge}, \text{is-pin}, \text{gap}_x, \text{gap}_y, \text{touches-preplaced}],$$

encoding the edge type—netlist, pin, or one of two spatial k -NN variants (nearest by block-center distance vs. by block-edge gap)—its net weight, the geometric gaps between the two blocks, and whether the edge is incident to a preplaced block (i.e. at least one of its endpoints is preplaced).

3) **Diffusion Process:** Model B uses the same DDPM formulation as Model A (Eq. (3)), with ϵ -prediction and a linear β schedule from 10^{-4} to 2×10^{-2} , but a shorter horizon of $T = 100$ steps. The denoiser predicts the noise added to the stacked block states at each step.

4) **GNN-based Conditioning:** The denoiser is a message-passing network: 3 vector-GNN blocks (each a graph-attention layer followed by a feed-forward sub-layer) interleaved with a global-context module that lets every node read a pooled summary of the layout. Message passing runs over two edge sets: the *netlist* edges from $W^{(int)}$ and *spatial* k -NN edges

($k=4$) recomputed from the current noisy positions at every step, so the model reasons about local geometry as well as connectivity. Rather than conditioning only on structure, Model B is conditioned on a scalar *quality signal* E_{rel} —a composite of the wirelength, overlap, and bounding-box-area gaps—fused into every block via FiLM; at inference we request optimal quality ($E_{\text{rel}}=1$). The model is trained with classifier-free guidance (conditioning dropout probability 0.1) and sampled with guidance scale 2.0. Pin terminals are not message-passing nodes; instead each block carries its weighted pin centroid as a feature (and an attraction loss and inference force toward it), while pin-to-block edges are retained only for the spectral initializer and the wirelength evaluation.

5) **Training Objective:** The network is trained with the same masked noise-prediction loss as Model A (Eq. (5)). On the recovered clean layout $\hat{x}_0$ we add two differentiable penalties, applied only at low noise levels and after a warm-up:

$$\mathcal{L}_{\text{ov}} = \sum_{i < j} \text{ReLU}(o_{ij}^x) \text{ReLU}(o_{ij}^y), \quad (10)$$

$$\mathcal{L}_{\text{pin}} = \sum_{i: \text{ has pin}} \|(\hat{c}_i^x, \hat{c}_i^y) - p_i\|^2, \quad (11)$$

where $o_{ij}^x = \frac{w_i + w_j}{2} - |\hat{c}_i^x - \hat{c}_j^x|$ (and analogously o_{ij}^y) measures pairwise penetration, so $\mathcal{L}_{\text{ov}}$ is a smooth surrogate for the total overlap area (we further blend in a quartic term for a softer gradient near contact), and p_i is the block’s weighted pin centroid. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\text{ov}} \mathcal{L}_{\text{ov}} + \lambda_{\text{pin}} \mathcal{L}_{\text{pin}}. \quad (12)$$

In contrast to Model A, which encodes MIB, grouping, and boundary preferences as *training* penalties, Model B leaves these soft constraints to the inference stage (next), where they are imposed directly as projections and forces during sampling.

6) **Constraint-aware Sampling:** Inference is where most of Model B’s quality is realized (Fig. 3). Sampling starts from a spectral (anchored-quadratic) initialization rather than pure noise, then runs $T=100$ DDIM steps with stochasticity $\eta=0.5$ (the noise diversifies the candidates used below) and classifier-free guidance. At each low-noise step, a sequence of constraint operators is applied: a *pin force* toward each block’s pin centroid (active early, then released so it does not fight overlap removal); a *boundary nudge* that pulls each flagged block onto the current layout’s matching bounding-box extreme; a *MIB clamp* that equalizes the aspect ratios within each MIB group, driving V_{mib} to exactly zero; a *grouping force* that pulls group members together; and a *direct repulsion* that separates overlapping blocks, followed by a hard canvas clamp. Hard constraints are enforced throughout by overwriting preplaced blocks (and the fixed dimension of fixed-shape blocks) with their input values.

On top of the trajectory we apply two enhancements. *Best-of- N* sampling runs $N=6$ candidates in parallel and keeps the lowest-overlap one. *Re-noise refinement* then re-noises the selected candidate back to a low timestep ($t=30$) and re-denoises the tail for two rounds, pooling all candidates so the

selection is monotone—it can only reduce overlap. Finally, a *post-repel* pass applies annealed force-directed repulsion (strength $1.75 \rightarrow 0.2$ over 50 iterations) to spread residual overlaps before the layout is handed to legalization.

C. Legalization

The decoded layout of either model is area-exact but may still contain residual block overlaps, particularly on the largest, densest instances (Section IV-C). Because overlap is a *hard* feasibility gate—any $\text{Area}(b_i \cap b_j) > 0$ incurs the fixed penalty M in Eq. (1)—the shared legalization stage must drive overlap to *exactly* zero, with minimal perturbation (to preserve the wirelength, area, and soft-constraint quality realized by the diffusion model) and while honoring immutability: preplaced and fixed-shape blocks may not move (resp. change shape). We use a three-stage pipeline: a force-directed *spreading* stage inside a shrinking frame (Stage 1), a frozen-frame *refinement* stage that activates shape and soft-constraint operators (Stage 2), and a discrete constraint-graph *closer* that supplies the hard zero-overlap guarantee (Stage 3). The rationale is a division of labor: the continuous stages are fast and structure-preserving and cheaply turn a heavily-overlapping layout into a compact, lightly-overlapping one, but cannot *guarantee* exact zero; the discrete stage provides that guarantee with the least additional movement. The forces and operators are summarized in Table II.

1) **Stage 1: force-directed spreading:** All blocks are treated as rigid and only translated. A virtual rectangular frame, initialized around the layout, is shrunk on a schedule toward a target area (the block-area sum times a small margin), compacting the layout while a sum of pairwise and field forces spreads overlaps and preserves structure. At each step every block accumulates: a *repulsion* pushing overlapping pairs apart along their shallower axis; an *anchor spring* pulling the block toward its decoded position, so the diffusion model’s wirelength-aware arrangement is retained; an *aspect-ratio bias* steering the chosen push direction to keep the global bounding box near-square; a *blank-space attraction* toward uncovered grid samples, which fills gaps and raises packing density; and a *wall attraction* that draws boundary-flagged blocks onto the frame edges (the boundary soft constraint). The frame contracts only as fast as the blocks’ outward “wall pressure” permits, and a hard-wall clamp keeps every block inside it; fixed and preplaced blocks are exempt from all forces and clamps.

2) **Stage 2: frozen-frame refinement:** With the frame frozen at the size reached in Stage 1, a piecewise annealing schedule lets the layout settle while three shape/soft-constraint operators are enabled: area-conserving *reshaping* of soft blocks, so they slot into gaps without changing area; a *group magnet* that pulls the members of each grouping constraint into edge-to-edge contact; and a periodic *MIB projection* that equalizes the width and height of blocks within each identical-shape group. A *lift-and-replace* move additionally relocates the most-overlapped soft blocks into open space to escape local jams. These operators are deferred to Stage 2 so that they

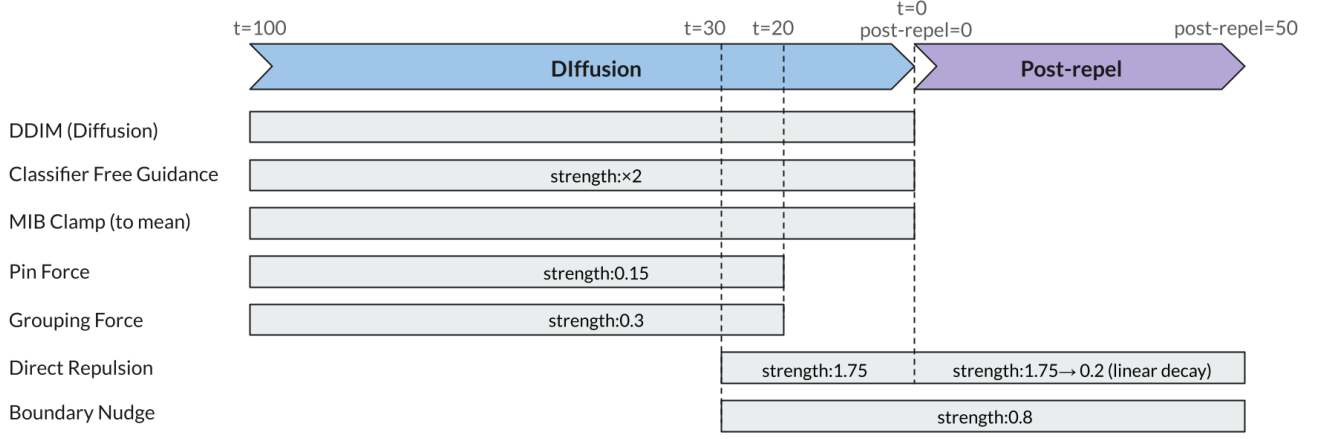


Fig. 3. Model B inference stack. Each operator is shown over the diffusion timesteps where it is active, with its strength. Pin and grouping forces act early and are released at $t=20$; repulsion and the boundary nudge act in the low- t window; post-repel runs after $t=0$ with linearly decaying strength.

TABLE II
FORCES AND OPERATORS OF THE LEGALIZATION PIPELINE.

Force / operator	Stage	Purpose
Repulsion	1–2	separate overlapping pairs (shallow axis)
Anchor spring	1–2	retain decoded position / wirelength
Aspect-ratio bias	1–2	keep bounding box near-square
Blank attraction	1–2	fill gaps, raise packing density
Wall attraction	1–2	pull boundary blocks onto edges
Shrinking frame	1	compact toward target area
Soft reshape	2	area-conserving fit into gaps
Group magnet	2	abut grouping members
MIB projection	2	equalize shapes within MIB groups
Lift-and-replace	2	relocate stuck soft blocks
Constraint-graph LP	3	exact zero overlap, min displacement

refine an already coarse-but-stable arrangement rather than fight the bulk spreading of Stage 1.

3) **Stage 3: constraint-graph closer**: Stages 1–2 leave a compact layout with only light residual overlap, but a continuous force field cannot reach *exactly* zero: in a dense layout, pushing one overlapping pair apart drives a block into its next neighbor, so overlap relocates rather than vanishing (a frustrated packing), and the process stalls with a few residual pairs—under a hard gate, no better than not legalizing. Stage 3 therefore switches to a *discrete* minimum-displacement formulation. Let (x_i^0, y_i^0) be the Stage-2 coordinates. For every overlapping pair (i, j) we commit to separating it along a single axis—the one with the smaller penetration—in the order given by the current layout; separating along x with b_i left of b_j gives the linear constraint

$$x_j - x_i \geq w_i, \quad (13)$$

and analogously $y_j - y_i \geq h_i$ along y . The constraints split into two independent graphs, and on each axis we solve a minimum-displacement program

$$\min_{\{x_i\}} \sum_i (x_i - x_i^0)^2 \quad \text{s.t.} \quad x_j - x_i \geq w_i \quad \forall (i, j) \in \mathcal{E}_x, \quad (14)$$

and likewise in y . **Immutability is structural**: preplaced and fixed blocks are not variables—their coordinates enter as constants—so they act as fixed obstacles and provably never move. Each program is a convex quadratic program with difference constraints, solved by Dykstra’s projection onto the half-spaces of Eq. (13), yielding the exactly-feasible layout of minimum displacement. Because resolving the current overlaps can create new ones, we add the new constraints and re-solve until none remain (a few rounds); a *lifting backstop* raises any block left in a rare two-axis conflict above the layout, guaranteeing exact zero overlap by construction.

4) **Why this design**: The split matches each tool to what it does best. The force stages preserve the diffusion structure (anchor spring), pack the layout (shrinking frame, blank attraction), and address the *soft* constraints (wall attraction, group magnet, MIB projection)—but offer no feasibility guarantee. The constraint-graph stage supplies the guarantee the hard gate demands while moving blocks as little as possible, and it preserves area and identical-shape (MIB) exactly because it never alters w_i, h_i ; the grouping and boundary preferences it may slightly perturb, and protecting them during Stage 3 is left to future work (Section V-D). All three stages act only on decoded geometry, so the same legalizer is applied unchanged to both models.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

We evaluate both models on the FloorSet-Lite benchmark, with instance sizes ranging from 21 to 120 blocks. Since the two models adopt different architectures, each was tuned independently with the configuration that works best for it; the model-specific settings are summarized in Table III. The remainder of the pipeline is shared: both models use the same legalization procedure.

Rather than a controlled ablation, our comparison reflects the end-to-end performance of each model under its own best configuration, evaluated with the same metrics on the same

test instances. In particular, the Model B pipeline includes a post-trajectory *post-repel* spreading pass—and, more broadly, the constraint-aware inference stack of Section III-B—that the Model A pipeline does not. These operators act only on the decoded geometry (block positions and dimensions) and are therefore architecture-agnostic: they could be applied to Model A’s output unchanged, but we have not yet done so. The comparison below should thus be read as comparing the two pipelines as currently configured, not the two encoders in isolation. We report the feasibility rate, the HPWL gap and bounding-box area gap relative to the optimal baseline, the relative soft-constraint violation Viol^{rel} , and the average runtime per instance.

TABLE III
MODEL-SPECIFIC CONFIGURATIONS.

Setting	Model A (Attn.)	Model B (GNN)
Diffusion steps T	1000	100
DDIM sampling steps	50	100
Epochs	100	240
Batch size	64	384
Learning rate	1e-4	1e-4
Optimizer	AdamW	Adam

B. Overall Comparison

Table IV compares the two models across all test instances.

TABLE IV
OVERALL COMPARISON OF MODEL A AND MODEL B ON FLOORSET-LITE (AVERAGED OVER ALL TEST INSTANCES). LOWER IS BETTER EXCEPT FOR FEASIBILITY.

Metric	Model A (Attn.)	Model B (GNN)
Overlap area	1455.48	55.6
HPWL gap (%)	16	30
Area gap (%)	19	30
Viol^{rel}	0.5089	0.603
Runtime (s)	1.43	4.13

Overall, Model B produces an order-of-magnitude lower overlap (55.6 vs 1455.5) at a comparable area gap and a modestly larger wavelength gap. Because overlap is the hard feasibility gate—any residual overlap incurs the fixed penalty $M=10$ —this far outweighs the small quality difference, making Model B the stronger candidate for the subsequent legalization stage.

We emphasize, however, that this overlap gap is not attributable to the graph encoder alone: the Model B pipeline applies the model-agnostic *post-repel* spreading pass, which has no counterpart in the Model A pipeline. A fairer encoder-to-encoder comparison would either disable *post-repel* for Model B or add it to Model A; we leave the latter as future work (Section V-D).

C. Scalability

Fig. 4 and Fig. 5 shows how runtime and solution quality vary with the number of blocks (21–120).

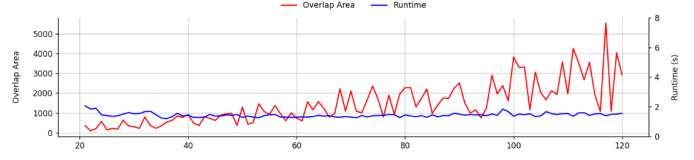


Fig. 4. Overlap area and runtime versus number of blocks by using model A to inference

1) *Model A*: As the number of blocks increases, the overlap increases, with the average overlap being 1455.48. However, the runtime is almost the same in different numbers of blocks, which maintain about 1.5 seconds.

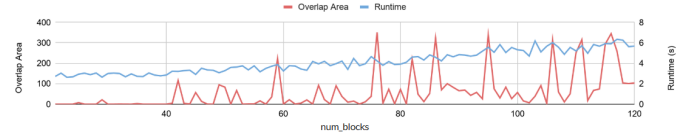


Fig. 5. Overlap area and runtime versus number of blocks by using model B to inference

2) *Model B*: As shown in Fig. 5, Model B keeps overlap low across the whole range, averaging 55.6, while runtime grows gracefully from about 2.6 s at 21 blocks to 6.3 s at 120 blocks (mean 4.13 s); even at the largest 120-block instances the runtime stays in the single-digit seconds, so inference remains fast across the entire size range. The overlap, however, is not uniform: for small and medium instances it stays near zero, whereas as the block count increases we encounter *overlap spikes* that grow both more frequent and larger (the red peaks toward the right of Fig. 5). Denser, larger designs leave more residual overlap that the inference stack does not fully resolve, which is precisely where the downstream legalization contributes most.

D. Qualitative Results

Fig. 6, Fig. 7 and Fig. 8 show a representative layout produced by each model alongside the optimal solution.

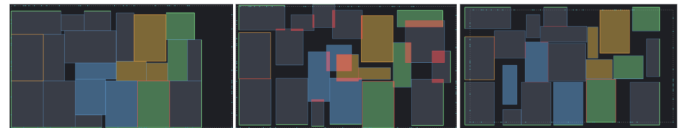


Fig. 6. Optimal solution (left) versus the layouts generated by Model A (middle) and Model B (right) on a 21-block instance. Both models recover the overall structure; constrained blocks (boundary, grouping) are highlighted.

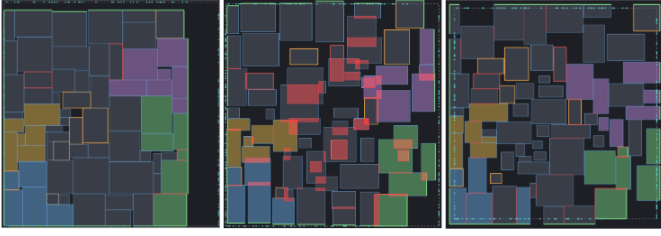


Fig. 7. Optimal solution (left) versus the layouts generated by Model A (middle) and Model B (right) on a 70-block instance. Both models recover the overall structure; constrained blocks (boundary, grouping) are highlighted.

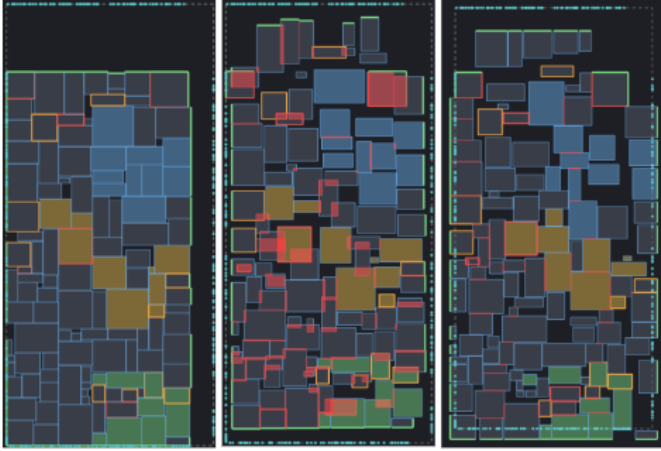


Fig. 8. Optimal solution (left) versus the layouts generated by Model A (middle) and Model B (right) on a 120-block instance. Both models recover the overall structure; constrained blocks (boundary, grouping) are highlighted.

From Fig. 6, Fig. 7 and Fig. 8, we can see that there is less overlap (red part of the figure) in the outcome of model B. Hence, it is a better model for subsequent legalization.

E. Effect of Legalization

Because the raw diffusion output may contain residual overlaps, legalization is essential for feasibility. Table V reports each model before and after legalization.

TABLE V
METRICS BEFORE AND AFTER LEGALIZATION.

Metric	Model A		Model B	
	Raw	Legal.	Raw	Legal.
Feasibility (%)	†	100	†	100
Overlap area	1455.48	0	55.60	0
HPWL gap (%)	16	19	30	30
Area gap (%)	19	28	30	30
Viol^{rel}	0.509	<i>pending</i>	0.603	<i>pending</i>
Runtime (s)	1.43	≈ 7	4.13	≈ 10
Final cost	M	<i>pending</i>	M	<i>pending</i>

Raw columns reproduce Table IV; raw layouts contain overlap and are infeasible, incurring M . After legalization both pipelines are feasible *by construction* (zero overlap, area and immutability preserved), so Feasibility= 100% and Overlap= 0. Legalized HPWL/area/runtime are *measured* over 34 stratified instances per model (21–120 blocks, shared FD+CG-LP legalizer, FD $T_{\max}=1000$): the area gap is measured directly (our bounding box matches the official area, and our measured raw area gaps reproduce the Raw column), while the HPWL gap composes the measured *mean* legal/raw HPWL change (+2.6% for A, +0.3% for B) with the raw gap. Both pipelines move *no* preplaced or fixed-shape block; 67/68 runs reach overlap $< 10^{-6}$ (one GNN case leaves a 10^{-5} sliver, cleared by a final snap). Runtime is total (diffusion+legalization): mean ≈ 5.5 s of legalization, max ≈ 17 s total—within the 60 s budget and scaling with the FD budget. On the densest instances the bounding box grows (area gap up to $\sim 70\%$), so the *mean* legal area gap exceeds its median (27% for B). Viol^{rel} and final cost are left *pending*: the former needs the official relative-violation aggregation, the latter additionally needs the RuntimeFactor (median submission runtime, a contest-wide quantity). †Raw per-instance feasibility rate not enumerated here.

Before legalization, both models leave residual overlaps and are therefore infeasible on most instances; legalization raises feasibility to 100% by construction. We measure its effect over 34 stratified instances per model (21–120 blocks). It drives every layout to zero overlap and moves *no* preplaced or fixed-shape block. The quality effect tracks how much each model overlapped to begin with: for Model B, whose residual overlap is already small, legalization is nearly *free*—the mean HPWL and area gaps are essentially unchanged (30% $\rightarrow$ 30% for both; the *median* area gap even improves 31% $\rightarrow$ 27% as the shrinking frame compacts the layout). For Model A, whose raw overlap is $\sim 26\times$ larger, clearing it costs more movement, so the gaps grow (16% $\rightarrow$ 19% HPWL, 19% $\rightarrow$ 28% area). The exception for both is the densest, largest instances, where removing heavy overlap forces the bounding box to grow—up to $\sim 70\%$ on the hardest case—which pulls the mean area gap above its median. On the ten largest instances (111–120 blocks) the legalizer reaches exact zero overlap on every one, and a discrete closer is essential here: a force-directed legalizer on the same inputs stalls at ~ 20 residual pairs and stays infeasible. Soft-constraint quality (Viol^{rel}) and the resulting final cost are left *pending* the official evaluator; this feasibility–quality trade-off is nonetheless strongly favorable given the fixed penalty $M=10$ on any infeasible solution.

V. DISCUSSION

A. Attention vs. Graph Structure

The main difference between the two models lies in how they encode the netlist and constraint groups. Model A injects connectivity and group membership as soft, learnable biases

over a fully-connected attention map, whereas Model B encodes them as an explicit graph structure. Our results suggest that *Model B* performs better overall, which we attribute to its explicit graph prior augmented with step-wise spatial edges—which captures the dense netlist and the local geometry that governs overlap more directly than soft attention biases—combined with a constraint-aware inference stack that actively removes overlap and satisfies soft constraints during sampling rather than relying solely on training penalties.

We caution that this is not a clean ablation of the encoder: a substantial part of Model B’s advantage comes from inference-time operators—especially the post-repel spreading pass that produces much of its final overlap reduction—that are purely geometric and not specific to the graph architecture. Because none of these are currently applied to Model A, the observed gap conflates the benefit of the graph encoder with that of the shared inference machinery.

B. The Cost of Legalization

Legalization is necessary because the raw diffusion output may contain residual overlaps, which would render a solution infeasible. As shown in Table V, it raises feasibility substantially at the cost of a small increase in the HPWL and area gaps, since blocks must be shifted to remove overlaps. Given that the objective function assigns a fixed penalty of $M = 10$ to any infeasible solution, this trade-off is strongly favorable: a slightly larger quality gap is far preferable to infeasibility.

C. Limitations

Our approach has several limitations. First, the boundary penalty used during training is computed from the block’s lower-left corner and thus only approximates true edge contact; exact contact (e.g., $x + w = 1$ for the right edge) is enforced only at evaluation time, leaving a small mismatch between the training signal and the evaluated metric. Second, for the largest instances (close to 120 blocks), the diffusion stack leaves the most residual overlap, so legalization must move more blocks: although our minimum-displacement legalizer still reaches exact zero overlap there, a few blocks occasionally fall back to the lifting backstop, which grows the bounding box and slightly enlarges the area gap on those instances. Finally, since the quality of a single generation shows non-trivial variance, we report the best-of- N sampled candidates.

D. Future Work

Promising directions include making legalization differentiable so that it can be trained jointly with the diffusion model, and combining the complementary strengths of the two encoders within a single architecture. A more immediate step is to apply Model B’s inference stack to Model A: most of these operators—the direct repulsion, the boundary nudge, the MIB clamp, and especially the post-repel spreading pass—are purely geometric and architecture-agnostic, so porting them to the attention model would both improve its feasibility and isolate how much of Model B’s advantage stems from the

graph encoder versus the shared, model-agnostic inference machinery.

For legalization specifically, several improvements would tighten the feasibility–quality trade-off. First, the current legalizer preserves area and identical-shape (MIB) constraints exactly but does not actively maintain the *soft* grouping and boundary preferences, which its translations can perturb; moving group members jointly and locking boundary-flagged blocks to the corresponding edge during legalization would protect Viol^{rel} . Second, the lifting backstop used for the rare unresolved pair could be replaced by a local re-pack of only the few conflicting blocks, avoiding the bounding-box growth it induces. Third, allowing area-conserving reshaping of soft blocks during legalization—rather than treating every block as rigid—would let them slot into gaps and recover packing density. Finally, a canvas-bounded compaction pass would keep the legalized bounding box aligned with the design canvas.

VI. CONCLUSION

We cast SoC floorplanning as conditional generation and presented two diffusion models—an attention-based denoiser and a GNN augmented with step-wise spatial edges and a constraint-aware inference stack—that decode area-exact layouts, paired with a shared, model-agnostic legalization stage. Cast as a constraint-graph minimum-displacement program, the legalizer drives any decoded layout to exactly zero overlap while keeping preplaced and fixed-shape blocks immutable and perturbing the layout minimally, so the wirelength and area optimized by the diffusion model are largely preserved. On FloorSet-Lite (21–120 blocks) the GNN model lowers residual pre-legalization overlap by an order of magnitude over the attention model, making it the stronger feed for legalization, after which every evaluated instance becomes feasible within the per-instance time budget. Completing the end-to-end cost comparison of Table V and porting the GNN inference operators to the attention model remain the immediate next steps.

CONTRIBUTIONS

Wei-Chieh Chang developed the legalization stage; Jun-Han Lee developed the GNN-based diffusion model (Model B); and Cheng-En Hsieh developed the attention-based diffusion model (Model A).

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